# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `test` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 2-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 48 023 254 960 | 3 820 802 773 | 3 797 752 332 | 22 956 427 | 94 014 | 99.4 | 21.32 | 0.10 | 15.764 | 64.32 | 4.453 | 3.914 | 0.126 | 0.159 |
| parest | 37 262 401 281 | 130 109 038 | 16 148 624 | 110 058 554 | 3 901 860 | 12.4 | 0.22 | 0.59 | 11.595 | 57.67 | 9.578 | 3.513 | 2.118 | 1.649 |
| deepsjeng | 46 202 686 169 | 1 453 401 396 | 380 604 124 | 1 046 384 737 | 26 412 535 | 26.2 | 2.78 | 4.53 | 9.980 | 6.89 | 2.859 | 7.613 | 3.282 | 3.898 |
| leela | 26 063 891 644 | 808 702 134 | 26 356 444 | 749 189 404 | 33 156 286 | 3.3 | 0.36 | 5.75 | 6.901 | 20.72 | 9.698 | 0.425 | 1.766 | 0.859 |
| namd | 31 914 751 099 | 336 702 115 | 334 454 030 | 1 939 340 | 308 745 | 99.3 | 3.13 | 0.01 | 12.887 | 25.71 | 5.486 | 1.313 | 3.844 | 3.891 |
| lbm | 6 506 800 040 | 274 131 044 | 274 130 500 | 544 | 0 | 100.0 | 9.67 | 0.00 | 23.735 | 88.75 | 0.424 | 2.261 | 15.074 | 0.184 |
| nab | 6 691 483 051 | 152 456 167 | 124 535 928 | 24 348 560 | 3 571 679 | 81.7 | 7.90 | 0.73 | 6.914 | 4.63 | 1.389 | 2.123 | 0.898 | 1.211 |
| xz | 2 014 351 521 | 34 435 119 | 4 683 669 | 26 392 608 | 3 358 842 | 13.6 | 0.68 | 2.62 | 6.496 | 25.90 | 0.124 | 1.002 | 2.222 | 2.241 |
| xz_4m_l1 | 2 001 462 938 | 30 569 523 | 3 959 703 | 23 548 312 | 3 061 508 | 13.0 | 0.59 | 2.35 | 7.996 | 35.15 | 0.121 | 0.808 | 2.863 | 1.815 |
| xz_4m_l2 | 2 083 103 943 | 29 428 092 | 3 700 891 | 22 776 791 | 2 950 410 | 12.6 | 0.54 | 2.19 | 9.797 | 42.07 | 0.116 | 0.814 | 3.396 | 1.908 |
| xz_4m_l3e | 10 573 516 663 | 134 644 921 | 61 575 948 | 66 069 008 | 6 999 965 | 45.7 | 2.51 | 1.25 | 6.916 | 19.19 | 1.914 | 1.167 | 1.747 | 3.242 |
| xz_4m_l4 | 4 011 967 585 | 68 464 242 | 20 213 033 | 44 411 548 | 3 839 661 | 29.5 | 2.12 | 2.21 | 8.369 | 30.25 | 2.255 | 2.394 | 1.558 | 2.249 |
| xz_4m_l4e | 12 115 711 352 | 137 799 962 | 63 036 496 | 67 532 844 | 7 230 622 | 45.7 | 2.28 | 1.11 | 6.702 | 17.71 | 2.020 | 1.170 | 1.810 | 3.475 |
| xz_1m_l0 | 711 271 503 | 13 362 560 | 1 091 347 | 10 938 543 | 1 332 670 | 8.2 | 0.62 | 3.08 | 5.939 | 29.24 | 0.168 | 2.012 | 1.021 | 1.382 |
| xz_1m_l1 | 719 848 327 | 12 739 426 | 973 549 | 10 458 508 | 1 307 369 | 7.6 | 0.58 | 2.91 | 6.727 | 33.78 | 0.168 | 1.706 | 1.215 | 1.441 |
| xz_1m_l2 | 724 690 222 | 12 498 311 | 956 442 | 10 241 945 | 1 299 924 | 7.7 | 0.57 | 2.83 | 6.928 | 34.23 | 0.148 | 1.718 | 1.254 | 1.484 |
| xz_1m_l3e | 1 994 406 562 | 34 952 797 | 12 876 553 | 20 010 020 | 2 066 224 | 36.8 | 2.59 | 2.01 | 6.214 | 18.66 | 1.543 | 2.131 | 0.775 | 2.798 |
| xz_1m_l4 | 1 076 057 785 | 23 230 841 | 5 156 845 | 16 486 476 | 1 587 520 | 22.2 | 2.01 | 3.06 | 6.832 | 24.25 | 1.730 | 4.355 | 0.697 | 2.351 |
| xz_1m_l4e | 2 198 787 666 | 35 410 889 | 13 089 895 | 20 222 586 | 2 098 408 | 37.0 | 2.43 | 1.84 | 6.173 | 17.52 | 1.620 | 2.109 | 0.800 | 2.919 |
| imagick | 117 733 556 | 2 444 490 | 2 450 | 2 437 082 | 4 958 | 0.1 | 0.02 | 4.14 | 4.422 | 81.36 | 10.069 | 69.568 | 1.302 | 0.348 |
| blender | 994 421 902 | 46 417 461 | 4 430 041 | 40 844 530 | 1 142 890 | 9.5 | 1.16 | 8.21 | 5.341 | 19.43 | 2.281 | 3.165 | 2.206 | 2.767 |
| povray | 2 240 986 547 | 168 867 726 | 2 121 674 | 155 851 654 | 10 894 398 | 1.3 | 0.27 | 13.91 | 10.145 | 41.21 | 7.406 | 20.677 | 1.842 | 1.807 |
| omnetpp | 12 426 267 350 | 842 780 888 | 5 233 033 | 791 025 406 | 46 522 449 | 0.6 | 0.09 | 12.73 | 12.565 | 26.59 | 0.250 | 0.863 | 1.933 | 2.478 |
| xalancbmk | 395 919 938 | 18 468 307 | 295 564 | 17 282 226 | 890 517 | 1.6 | 0.23 | 8.73 | 13.661 | 33.74 | 9.999 | 18.696 | 3.437 | 4.325 |
| x264 | 475 535 613 470 | 19 842 562 937 | 13 777 288 367 | 5 322 257 634 | 743 016 936 | 69.4 | 6.24 | 2.24 | 2.326 | 26.97 | 0.389 | 0.954 | 1.487 | 1.610 |
| gcc | 16 026 850 | 1 021 294 | 52 739 | 952 288 | 16 267 | 5.2 | 0.84 | 11.88 | 7.167 | 32.52 | 0.666 | 2.169 | 1.438 | 1.209 |
| mcf | 24 426 982 047 | 234 360 896 | 91 410 910 | 142 147 076 | 802 910 | 39.0 | 1.58 | 1.16 | 25.499 | 75.88 | 5.567 | 8.282 | 3.676 | 5.825 |
| perlbench_makerand | 2 400 742 243 | 222 681 834 | 55 | 222 662 746 | 19 033 | 0.0 | 0.08 | 18.55 | 5.716 | 0.00 | 0.000 | 2.725 | 0.879 | 0.880 |
| bwaves_1 | 17 288 358 363 | 145 483 800 | 145 451 376 | 26 481 | 5 943 | 100.0 | 3.55 | 0.00 | 41.067 | 31.22 | 2.704 | 0.330 | 4.063 | 5.189 |
| bwaves_2 | 17 287 430 556 | 145 483 720 | 145 451 376 | 26 421 | 5 923 | 100.0 | 3.55 | 0.00 | 41.069 | 31.22 | 2.704 | 0.330 | 4.069 | 5.193 |
| exchange2 | 57 667 143 431 | 1 874 880 445 | 1 304 771 747 | 345 493 349 | 224 615 349 | 69.6 | 5.03 | 1.20 | 2.504 | 1.08 | 2.742 | 3.560 | 7.229 | 4.833 |
| fotonik3d | 3 098 360 920 | 28 397 309 | 27 853 226 | 437 206 | 106 877 | 98.1 | 3.84 | 0.03 | 17.861 | 41.35 | 5.709 | 0.057 | 1.160 | 3.747 |
| roms | 28 955 766 090 | 203 007 943 | 182 777 748 | 12 238 510 | 7 991 685 | 90.0 | 4.79 | 0.08 | 20.998 | 38.82 | 4.438 | 1.564 | 1.413 | 2.838 |
| wrf | 84 396 898 728 | 420 514 578 | 109 794 207 | 285 402 467 | 25 317 904 | 26.1 | 0.91 | 0.68 | 18.055 | 22.28 | 9.398 | 6.610 | 0.844 | 1.636 |
| cam4 | 33 466 803 539 | 585 237 550 | 168 288 974 | 404 426 599 | 12 521 977 | 28.8 | 2.11 | 2.42 | 10.243 | 52.99 | 2.656 | 2.913 | 1.165 | 1.702 |
| **Total** | 1 003 605 899 841 | 32 326 452 528 | **21 110 519 840** | 10 037 478 430 | 1 178 454 258 | 65.3 | 5.17 | 2.00 | 0.000 | — | — | — | — | — |